Pizza Sales SQL Analysis

Project Summary

This report presents SQL-based analysis performed on a pizza sales dataset to generate business insights such as revenue trends, customer ordering behavior, and product performance.

Objective

The goal of this project is to demonstrate practical SQL skills by solving real business questions using:

- Joins
- Aggregations
- Subqueries
- Window Functions
- KPI analysis

Dataset Description

The dataset contains four relational tables:

1. orders

- order_id
- date
- time

2. order_details

- order_details_id
- order_id
- pizza_id
- quantity

3. pizzas

- pizza_id
- pizza_type_id
- size
- price

4. pizza_types

- pizza_type_id
- name
- category
- ingredients

Database Relationships

- orders → order_details (1 to many)
- pizzas → order_details (1 to many)
- pizza_types → pizzas (1 to many)

These relationships allow full sales analysis from order level to product level.

Business Questions Solved

1. Total Revenue

Calculated total revenue generated from pizza sales.

2. Highest Priced Pizza

Identified the pizza with the maximum price.

3. Most Common Pizza Size

Determined which pizza size customers order most.

4. Quantity Sold by Category

Analyzed how many pizzas were sold in each category.

5. Orders by Hour

Studied order distribution throughout the day.

6. Top 3 Pizzas by Revenue

Identified the top performing pizzas based on revenue.

7. Revenue Contribution

Calculated percentage contribution of each category to total revenue.

8. Cumulative Revenue Over Time

Analyzed revenue growth using window functions.

9. Top 3 Pizzas per Category

Used ranking window function to find top products per category.

SQL Skills Demonstrated

- Complex Joins
- Aggregate Functions
- Subqueries
- Window Functions (RANK)
- Data Aggregation
- Business KPI Analysis

Key Insights

- Identified peak ordering hours.
- Found high revenue generating pizza types.
- Discovered best performing pizza categories.
- Analyzed customer ordering patterns.

Conclusion

This project demonstrates strong SQL skills and the ability to convert raw data into meaningful business insights.